NLPNN-Based Constrained MPC for a QuadCopter

1. Quad Modelling

The quadcopter state and control vectors are

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{r}^T & \boldsymbol{v}^T & \boldsymbol{\eta}^T & \boldsymbol{\omega}^T \end{bmatrix}^T \in \mathbb{R}^{12}, \qquad \boldsymbol{u} = \begin{bmatrix} T & \tau_\phi & \tau_\theta & \tau_\psi \end{bmatrix}^T,$$

where $\boldsymbol{r} = [x, y, z]^T$, $\boldsymbol{v} = [v_x, v_y, v_z]^T$, $\boldsymbol{\eta} = [\phi, \theta, \psi]^T$, and $\boldsymbol{\omega} = [p, q, r]^T$. Using the Newton–Euler model,

$$\dot{\boldsymbol{r}} = \boldsymbol{v}, \tag{1}$$

$$\dot{\boldsymbol{v}} = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} + R(\boldsymbol{\eta}) \begin{bmatrix} 0 \\ 0 \\ T/m \end{bmatrix}, \tag{2}$$

$$\dot{\boldsymbol{\eta}} = E(\phi, \theta)^{-1} \boldsymbol{\omega}, \tag{3}$$

$$\dot{\boldsymbol{\omega}} = J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega}). \tag{4}$$

Here $R(\boldsymbol{\eta})$ rotates body-frame thrust into the inertial frame, $J = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})$, and $\boldsymbol{\tau} = [\tau_\phi, \tau_\theta, \tau_\psi]^T$.

The quadcopter parameters used in simulation are

$$m = 0.468 \text{ kg}, \qquad g = 9.81 \text{ m/s}^2, \qquad l = 0.17 \text{ m},$$

$$J = \mathrm{diag}(0.0023,\ 0.0023,\ 0.0046) \text{ kg m}^2.$$

The MPC prediction model is intentionally mismatched, using $m_{\text{nom}} = 0.440$ kg and

$$J_{\text{nom}} = \mathrm{diag}(0.0020,\ 0.0026,\ 0.0041) \text{ kg m}^2,$$

so that the NLPNN learns the residual dynamics between the nominal and actual models.

2. NN Observer Structure

The adaptive prediction model is written as

$$\dot{\hat{\boldsymbol{x}}}_p = f_{\text{nom}}(\hat{\boldsymbol{x}}, \boldsymbol{u}) + \hat{W}\, \sigma\left(\hat{V}^T \bar{\boldsymbol{x}}\right), \qquad \sigma(\cdot) = \tanh(\cdot),$$

with the scaled augmented input

$$\bar{\boldsymbol{x}} = \frac{[\hat{\boldsymbol{x}}^T\ \boldsymbol{u}^T]^T}{\boldsymbol{s}}.$$

The observer adds output-error injection,

$$\dot{\hat{\boldsymbol{x}}} = f_{\text{nom}}(\hat{\boldsymbol{x}}, \boldsymbol{u}) + \hat{W}\sigma(\hat{V}^T\bar{\boldsymbol{x}}) + G(\boldsymbol{x} - \hat{\boldsymbol{x}}),$$

where the implementation uses full-state output feedback and $G = 90I$. For online identification, define the residual target and model error as

$$\Delta f = f_{\text{true}}(\boldsymbol{x}, \boldsymbol{u}) - f_{\text{nom}}(\boldsymbol{x}, \boldsymbol{u}), \qquad \boldsymbol{e}_m = \Delta f - \hat{W}\boldsymbol{\sigma}.$$

The nonlinear-in-parameter weights are adapted using

$$\dot{\hat{W}} = \eta_1 \boldsymbol{e}_m \boldsymbol{\sigma}^T - \rho_1 \hat{W}, \tag{5}$$

$$\dot{\hat{V}} = \eta_2 \bar{\boldsymbol{x}} \left[\left(\hat{W}^T \boldsymbol{e}_m\right) \odot \left(1 - \boldsymbol{\sigma}^2\right)\right]^T - \rho_2 \hat{V}. \tag{6}$$

Thus both network layers adapt, while the ρ_i terms provide weight damping. The MPC uses the learned model $f_{\text{nom}} + f_{\text{NN}}$ for prediction; the observer correction term is not propagated into future predictions.

3. MPC Formulation, Cost Function and Constraints

At each control update, the learned nonlinear model is locally linearized about the current estimate and previous input:

$$\dot{\boldsymbol{x}} \approx A_c \boldsymbol{x} + B_c \boldsymbol{u} + \boldsymbol{c}_c.$$

With sampling time $T_s = 0.05$ s, forward-Euler discretization gives

$$\boldsymbol{x}_{k+1} = A_d \boldsymbol{x}_k + B_d \boldsymbol{u}_k + \boldsymbol{c}_d, \qquad A_d = I + T_s A_c, \qquad B_d = T_s B_c.$$

A prediction/control horizon of $N = 20$ is used.

The quadratic objective is

$$J = \sum_{k=0}^{N-1} \Big[(\boldsymbol{x}_{k+1} - \boldsymbol{x}_{r,k+1})^T Q (\boldsymbol{x}_{k+1} - \boldsymbol{x}_{r,k+1}) \tag{7}$$

$$+ (\boldsymbol{u}_k - \boldsymbol{u}_h)^T R (\boldsymbol{u}_k - \boldsymbol{u}_h) + \Delta \boldsymbol{u}_k^T S \Delta \boldsymbol{u}_k \Big] \tag{8}$$

$$+ (\boldsymbol{x}_N - \boldsymbol{x}_{r,N})^T P (\boldsymbol{x}_N - \boldsymbol{x}_{r,N}), \tag{9}$$

where $\boldsymbol{u}_h$ is the nominal hover input and $P = 4Q$. After condensing, this becomes the constrained QP

$$\min_{\boldsymbol{U}} \ \frac{1}{2} \boldsymbol{U}^T H \boldsymbol{U} + \boldsymbol{f}^T \boldsymbol{U} \qquad \text{subject to} \qquad \boldsymbol{l} \le A \boldsymbol{U} \le \boldsymbol{u}.$$

The implemented constraints are physically meaningful:

- **Thrust and moments:**

$$1.5 \le T \le 8.0 \text{ N}, \qquad |\tau_\phi|, |\tau_\theta| \le 0.040 \text{ N m}, \qquad |\tau_\psi| \le 0.028 \text{ N m}.$$

- **Input-rate limits per control step:**

$$|\Delta \boldsymbol{u}| \le \begin{bmatrix} 1.0 & 0.025 & 0.025 & 0.012 \end{bmatrix}^T.$$

- **Altitude and translational velocity:**

$$0.5 \le z \le 4.5 \text{ m}, \qquad |v_x|, |v_y| \le 2.5 \text{ m/s}, \qquad |v_z| \le 2.0 \text{ m/s}.$$

- **Attitude and body rates:**

$$|\phi|, |\theta| \le 30^\circ, \qquad |p|, |q| \le 3.0 \text{ rad/s}, \qquad |r| \le 2.2 \text{ rad/s}.$$

A small tightening margin is applied to the body-rate bounds to provide additional robustness against local-linearization and model mismatch.